

Regression Analysis

MATH 3424

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- Covariance and Correlation

Covariance of X, Y : $\text{Cov}(Y, X) = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{n-1} \begin{cases} > 0 & Y \sim X \text{ positive relationship} \\ < 0 & Y \sim X \text{ negative relationship} \end{cases}$

Let $s_y = \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}}$, $s_x = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$. By standardizing X, Y , we get

$$\text{Cor}(Y, X) = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{y_i - \bar{y}}{s_y} \cdot \frac{x_i - \bar{x}}{s_x} \right) = \frac{\text{Cov}(Y, X)}{s_y s_x} = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (x_i - \bar{x})^2}} \in [-1, 1]$$

- Simple Linear Regression

Construct a linear model $Y = \beta_0 + \beta_1 X + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$

We need to minimize $S(\beta_0, \beta_1) = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$, where $e_i = y_i - \hat{y}_i$ are called residuals.

The regression line is given by $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$, and for each observation we compute $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.

An unbiased estimate of σ^2 is given by $\hat{\sigma}^2 = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2} = \frac{\text{SSE}}{n-2}$

The values of $\hat{\beta}_0$ and $\hat{\beta}_1$ are given by $\hat{\beta}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$, and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

Equivalently, $\hat{\beta}_1 = \frac{\text{Cov}(Y, X)}{\text{Var}(X)} = \text{Cor}(Y, X) \frac{s_y}{s_x}$

Since $\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ are random, hence $\hat{\beta}_0$ and $\hat{\beta}_1$ are also random, following the distribution:

$$\hat{\beta}_0 \sim \mathcal{N}\left(\beta_0, \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]\right) \quad \hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)$$

$$\text{s.e.}(\hat{\beta}_0) = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} \quad \text{s.e.}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

```
> df = read.table("path/to/file", header = TRUE)
> model = lm(Y ~ X, data = df)
> plot(Y ~ X, data = df, xlab, ylab, main, pch, cex, col)
> abline(model, lwd = , col = )
> summary(model)
```

- Inference on Coefficients

Now we want to perform statistical analysis on $\hat{\beta}_0$ and $\hat{\beta}_1$.

$$\begin{cases} H_0: \beta_0 = \beta_0^0 \\ H_1: \beta_0 \neq \beta_0^0 \end{cases} \text{ we evaluate } t_0 = \frac{\hat{\beta}_0 - \beta_0^0}{\text{s.e.}(\hat{\beta}_0)} \sim t_{n-2} \quad \begin{cases} H_0: \beta_1 = \beta_1^0 \\ H_1: \beta_1 \neq \beta_1^0 \end{cases} \text{ we evaluate } t_1 = \frac{\hat{\beta}_1 - \beta_1^0}{\text{s.e.}(\hat{\beta}_1)} \sim t_{n-2}$$

H_0 is rejected if $|t| \geq t_{n-2, \frac{\alpha}{2}}$, or equivalently, $P(T_{n-2} \geq |t|) \leq \frac{\alpha}{2}$, $T_{n-2} \sim t_{n-2}$.

Alternatively, we can use the population correlation coefficient ρ , $\begin{cases} H_0: \rho = 0 \\ H_1: \rho \neq 0 \end{cases} \iff \begin{cases} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{cases}$

Test $t_1 = \frac{\text{Cor}(Y, X) \sqrt{n-2}}{\sqrt{1 - (\text{Cor}(Y, X))^2}} \sim t_{n-2}$. Reject when $|t_1| \geq t_{n-2, \frac{\alpha}{2}}$.

Confidence interval of $\hat{\beta}_0$ and $\hat{\beta}_1$ is given by $\hat{\beta}_0 \pm t_{n-2, \frac{\alpha}{2}} \cdot \text{s.e.}(\hat{\beta}_0)$ and $\hat{\beta}_1 \pm t_{n-2, \frac{\alpha}{2}} \cdot \text{s.e.}(\hat{\beta}_1)$

```
> confint(model, level = ...)
```

- Prediction

```
> new_data = data.frame(X = ...)
> predict(model, newdata = new_data)
```

First case: Prediction y_0 of the value of the response value Y . (Predict $\beta_0 + \beta_1 x_0 + \varepsilon$)

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0, \quad \text{s.e.}(\hat{y}_0) = \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}, \text{ hence prediction interval: } \hat{y}_0 \pm t_{n-2, \frac{\alpha}{2}} \cdot \text{s.e.}(\hat{y}_0)$$

```
> predict(model, newdata = new_data, interval = c("prediction"), level)
```

Second case: Estimation of the mean response μ_0 . (Predict $\beta_0 + \beta_1 x_0$)

$$\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0, \quad \text{s.e.}(\hat{\mu}_0) = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}, \text{ hence confidence interval: } \hat{\mu}_0 \pm t_{n-2, \frac{\alpha}{2}} \cdot \text{s.e.}(\hat{\mu}_0)$$

```
> predict(model, newdata = new_data, interval = c("confidence"), level)
```

- Mesuring the Quality of Fit

$$\text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2 = \text{SSR} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2$$

Orthogonality between $\hat{\mathbf{Y}}$ and $\hat{\mathbf{Y}} - \mathbf{Y}$: $\sum_{i=1}^n \hat{y}_i (\hat{y}_i - y_i) = 0$

Fundamental identity: $\text{SST} = \text{SSR} + \text{SSE}$

Property: $\text{Cor}(Y, \hat{Y}) = |\text{Cor}(Y, X)|$

Denote $R^2 = (\text{Cor}(Y, X))^2 = (\text{Cor}(Y, \hat{Y}))^2$, we have $R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}$

- Regression Line Through the Origin

Special case of simple linear model: no-intercept model $Y = \beta_1 X + \varepsilon$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2}, \quad \hat{y}_i = \hat{\beta}_1 x_i, \quad \hat{\sigma} = \sqrt{\frac{\text{SSE}}{n-1}}, \quad \text{s.e.}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^n x_i^2}}$$

The fundamental identity becomes $\sum_{i=1}^n y_i^2 = \sum_{i=1}^n \hat{y}_i^2 + \sum_{i=1}^n e_i^2$. We have $R^2 = \frac{\sum_{i=1}^n \hat{y}_i^2}{\sum_{i=1}^n y_i^2} = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n y_i^2}$

- Multiple Linear Regression

The data consist of n observations on a response variable Y and p predictors $X_1, X_2, \dots, X_p$.

The relationship between Y and $X_1, X_2, \dots, X_p$ is formulated as $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$.

$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i, i = 1, 2, \dots, n$

$$\text{Define } \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}, \boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}, \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}, \text{ we have } \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

By minimizing $S(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 \iff \nabla_{\boldsymbol{\beta}} S(\boldsymbol{\beta}) = 2\mathbf{X}^T(\mathbf{X}\boldsymbol{\beta} - \mathbf{y}) = \mathbf{0}$, we get $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$.

$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_p x_{ip}, i = 1, 2, \dots, n$, or equivalently, $\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$

$\hat{\sigma}^2 = \frac{\text{SSE}}{n - p - 1}$ is an unbiased estimate of σ^2 , where $\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2$

No-intercept model: $Y = \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$

- Centering and Scaling

Unit-length scaling:
$$\begin{cases} \tilde{Z}_y = \frac{Y - \bar{y}}{L_y} & \text{where } L_y = \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \\ \tilde{Z}_{x_j} = \frac{X_j - \bar{x}_j}{L_{x_j}} & \text{where } L_{x_j} = \sqrt{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2} \end{cases} \quad (\text{will have means 0 and length 1})$$

Standardizing:
$$\begin{cases} \tilde{Y} = \frac{Y - \bar{y}}{s_y} & \text{where } s_y = \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n - 1}} \\ \tilde{X}_j = \frac{X_j - \bar{x}_j}{s_{x_j}} & \text{where } s_{x_j} = \sqrt{\frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}{n - 1}} \end{cases} \quad (\text{will have means 0 and s.d. 1})$$

Recover $\hat{\beta}$ from $\hat{\theta}$ after OLS regression on $\tilde{Y} \sim \tilde{X}_1, \dots, \tilde{X}_p$:
$$\begin{cases} \hat{\beta}_j = \frac{s_y}{s_{x_j}} \hat{\theta}_j & j = 1, \dots, p \\ \hat{\beta}_0 = \bar{y} - \sum_{j=1}^p \hat{\beta}_j \bar{x}_j \end{cases}$$

```
new_data <- apply(data, 2, function(x) {(x - mean(x)) / sd(x)})
```

- Properties of Least Square Estimation

$$W = \frac{\text{SSE}}{\sigma^2} \sim \chi_{n-p-1}^2$$

Denote $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1} = (c_{ij})_{i,j=1}^p \in \mathbb{R}^{p \times p}$

We have $\text{Cov}(\hat{\beta}) = \sigma^2 \mathbf{C} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$. In other word, $\text{Var}(\hat{\beta}_j) = \sigma^2 c_{jj}$, $\text{Cov}(\hat{\beta}_i, \hat{\beta}_j) = \sigma^2 c_{ij}$

$\hat{\beta}_j$ follows the distribution $\hat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma^2 c_{jj})$

- Measuring the Quality of Fit

More variables will always yield a larger R^2 , which is not desired. We introduce R_a^2 as a fair measure.

$$R_a^2 = 1 - \frac{\text{SSE}/(n - p - 1)}{\text{SST}/(n - 1)} = 1 - \frac{n - 1}{n - p - 1} (1 - R^2)$$

- Inference on Coefficients

Now we want to test β_j :
$$\begin{cases} H_0 : \beta_j = \beta_j^0 \\ H_1 : \beta_j \neq \beta_j^0 \end{cases}$$
 We test $t_j = \frac{\hat{\beta}_j - \beta_j^0}{\text{s.e.}(\hat{\beta}_j)} \sim t_{n-p-1}$, where $\text{s.e.}(\hat{\beta}_j) = \hat{\sigma} \sqrt{c_{jj}}$.

Rejection region: $|t_j| \geq t_{n-p-1, \frac{\alpha}{2}}$

Confidence interval: $\hat{\beta}_j \pm t_{n-p-1, \frac{\alpha}{2}} \cdot \text{s.e.}(\hat{\beta}_j)$

- Hypothesis Testing on Linear Model

$$\begin{cases} H_0 : \text{Reduced model is adequate} \\ H_1 : \text{Full model is adequate} \end{cases}$$

Denote $\hat{y}_i$ and $\hat{y}_i^*$ as the predicted values of y_i by the full and reduced model, respectively.

Full model: $p + 1$ distinct parameters, $\text{SSE}_{\text{FM}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{H}) \mathbf{y}$, where $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$

Reduced model: k distinct parameters, $\text{SSE}_{\text{RM}} = \sum_{i=1}^n (y_i - \hat{y}_i^*)^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{H}_1) \mathbf{y}$, where $\mathbf{H}_1 = \mathbf{X}_1(\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top$

where $\mathbf{X}_1$ is $\mathbf{X}$ with only the k columns (corresponding to the variables retained) retained.

$$F = \frac{(\text{SSE}_{\text{RM}} - \text{SSE}_{\text{FM}})/(p + 1 - k)}{\text{SSE}_{\text{FM}}/(n - p - 1)} \sim F_{p+1-k, n-p-1}, \text{ reject } H_0 \text{ when } F \geq F_{p+1-k, n-p-1, \alpha}$$

Assume the reduced model has q variables, we have $F = \frac{(R_p^2 + R_q^2)/(p - q)}{(1 - R_p^2)/(n - p - 1)}$, $\text{df} = p - q$ and $n - p - 1$.

```
> full_model <- lm(Y ~ ., data = df)
> reduced_model <- lm(Y ~ X1 + X2, data = df)
> anova(reduced_model, full_model)
```

- Hypothesis Testing on the Overall Model

$$\begin{cases} H_0 : Y = \beta_0 + \varepsilon \\ H_1 : Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon \end{cases}$$

$$F = \frac{(\text{SSE}_{\text{RM}} - \text{SSE}_{\text{FM}})/(p + 1 - k)}{\text{SSE}_{\text{FM}}/(n - p - 1)} = \frac{(\text{SST} - \text{SSE})/p}{\text{SSE}/(n - p - 1)} = \frac{\text{SSR}/p}{\text{SSE}/(n - p - 1)} = \frac{\text{MSR}}{\text{MSE}} \sim F_{p, n-p-1}$$

$$\text{Denote } R_p = \text{Cor}(Y, \hat{Y}). \text{ We have } F = \frac{R_p^2/p}{(1 - R_p^2)/(n - p - 1)} \sim F_{p, n-p-1}$$

When the reduced model has only one coefficient less than the full model, say β_j , then $F = t_j^2$.

In other word, F-value with 1 and $n - p - 1$ degrees of freedom is equal to the square of t-value with $n - p - 1$ degrees of freedom.

```
> full_model <- lm(Y ~ ., data = df)
> reduced_model <- lm(Y ~ 1, data = df)
> anova(reduced_model, full_model)
```

- Prediction

Predicted value corresponding to $\mathbf{x}_0$ is given by $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_{01} + \dots + \hat{\beta}_p x_{0p}$.

Confidence interval for $\hat{y}_0$: $\hat{y}_0 \pm t_{n-p-1, \frac{\alpha}{2}} \cdot \text{s.e.}(\hat{y}_0)$, where $\text{s.e.}(\hat{y}_0) = \hat{\sigma} \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}$

Confidence interval for $\hat{\mu}_0$: $\hat{\mu}_0 \pm t_{n-p-1, \frac{\alpha}{2}} \cdot \text{s.e.}(\hat{\mu}_0)$, where $\text{s.e.}(\hat{\mu}_0) = \hat{\sigma} \sqrt{\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}$

- The Standard Regression Assumptions

1. Assumptions about the model:

Linearity ($Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$), determined by examining the scatter plot.

2. Assumptions about the errors $\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$

- | | |
|------------------------------------|--|
| (i) Normality | $\varepsilon_i \sim \mathcal{N}$ |
| (ii) Errors have mean zero | $\mu(\varepsilon_i) = 0$ |
| (iii) Constant variance assumption | $\text{Var}(\varepsilon_i) = \sigma$ |
| (iv) Independent-errors assumption | $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0, \forall i \neq j$ |

3. Assumptions about the predictors

- (i) $X_1, X_2, \dots, X_p$ are non-random.
- (ii) The values $x_{1j}, x_{2j}, \dots, x_{nj}$ are measured without error.
- (iii) $X_1, X_2, \dots, X_p$ are linearly independent from each other, so that $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}$ exists.

4. Assumptions about the observations

All observations are equally reliable.

- Leverage

The fitted values can also be written in the form $\hat{y}_i = p_{i1}y_1 + p_{i2}y_2 + \dots + p_{in}y_n$

Projection matrix $\mathbf{P} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = (p_{ij})_{i,j=1}^n \in \mathbb{R}^{n \times n}$, where $p_{ij} = \frac{1}{n} + \frac{(x_i - \bar{x})(x_j - \bar{x})}{\sum_{k=1}^n (x_k - \bar{x})^2}$

The leverage value of the i -th observation $p_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{k=1}^n (x_k - \bar{x})^2}$

The residuals $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{P})\mathbf{y}$, therefore $\text{Cov}(\mathbf{e}) = \text{Cov}((\mathbf{I} - \mathbf{P})\mathbf{y}) = \sigma^2(\mathbf{I} - \mathbf{P})$.

We have $\text{Var}(e_i) = \sigma^2(1 - p_{ii})$

- Standardized Residuals

To overcome unequal variances, we want to standardize e_i .

Internally standardized residual $r_i = \frac{e_i}{\hat{\sigma} \sqrt{1 - p_{ii}}} \sim \mathcal{N}(0,1)$, where $\hat{\sigma}^2 = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p - 1} = \frac{\text{SSE}}{n - p - 1}$

Externally standardized residual $r_i^* = \frac{e_i}{\hat{\sigma}_{(i)} \sqrt{1 - p_{ii}}} \sim t_{n-p-2}$, where $\hat{\sigma}_{(i)}^2 = \frac{(\sum_{k=1}^n e_k^2) - e_i^2}{n - p - 2} = \frac{\text{SSE}_{(i)}}{n - p - 2}$

by fitting the model to $n - 1$ observations omitting the i -th observation. We have $r_i^* = r_i \sqrt{\frac{n - p - 2}{n - p - 1 - r_i^2}}$

$r_1, r_2, \dots, r_n$ are uncorrelated with $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ or $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n$.

- Checking Linearity and Normality Assumptions

```
> model <- lm(Y ~ X, data = df)
> hist(X) > dotplot(X) > boxplot(X)
> plot(data$X, data$Y, xlab = "X1", ylab = "Y")
> abline(model)
```

Sort the standardized residuals $r_1, r_2, \dots, r_n$ from smallest to largest and write them as $r_{(1)}, r_{(2)}, \dots, r_{(n)}$.

Theoretical quantiles $z_{(j-0.5)/n} = \Phi^{-1}\left(\frac{j - 0.5}{n}\right)$

The normal Q-Q plot is the scatter plot of $(r_{(j)}, z_{(j-0.5)/n})_{j=1}^n$

```
> qqnorm(model$residuals)
> qqline(model$residuals)
```

Also, we can check the scatter plot of (x_j, r_j) or the scatter plot of $(\hat{y}_j, r_j)$.

```
> plot(model$fitted.values, model$residuals)
```

- Influence and Outlier Analysis

The average leverage is $\frac{\sum_{i=1}^n p_{ii}}{n} = \frac{p + 1}{n}$

The i -th data point has high leverage if $p_{ii} \geq 2 \left(\frac{\sum_{i=1}^n p_{ii}}{n} \right) = 2 \left(\frac{p + 1}{n} \right)$

Outlier in Y : Check if $r_i \geq 3$ Outlier in X_i : Check if $p_{ii} \geq 2 \left(\frac{p + 1}{n} \right)$

We can derive that $p_{ii} + \frac{e_i^2}{\text{SSE}} \leq 1$.

Let $\hat{\beta}_{0(i)}, \hat{\beta}_{1(i)}, \dots, \hat{\beta}_{p(i)}, \hat{y}_{0(i)}, \hat{y}_{1(i)}, \dots, \hat{y}_{n(i)}$ and $\hat{\sigma}_{(i)}^2$ be the values obtained with the i -th observation deleted.

```
> library(olsrr)
```

1. Cook's distance: $C_i = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(i)})^2}{\hat{\sigma}^2(p+1)} = \frac{r_i^2}{p+1} \cdot \frac{p_{ii}}{1-p_{ii}}$
Considered influential if $C_i > 1$.

```
> plot(1:n, cooks.distance(full_model))
> ols_plot_cooksd_bar(full_model)
```

2. Welsch and Kuh Measure: $DFITS_i = \frac{\hat{y}_i - \hat{y}_{i(i)}}{\hat{\sigma}_{(i)} \sqrt{p_{ii}}} = r_i^* \sqrt{\frac{p_{ii}}{1-p_{ii}}}$ (corresponds to $\sqrt{C_i}$)
Considered influential if $|DFITS_i| > 2 \sqrt{\frac{p+1}{n-p-1}}$.

```
> ols_plot_dffits(full_model)
```

3. Hadi's influence measure: $H_i = \frac{p_{ii}}{1-p_{ii}} + \frac{p+1}{1-p_{ii}} \cdot \frac{d_i^2}{1-d_i^2}$, where $d_i = \frac{e_i}{\sqrt{SSE}}$ is called normalized residual,
 $\frac{p_{ii}}{1-p_{ii}}$ is called the potential function, and $\frac{p+1}{1-p_{ii}} \cdot \frac{d_i^2}{1-d_i^2}$ is called the residual function.
The potential-residual plot is the scatter plot of $(\frac{p_{ii}}{1-p_{ii}}, \frac{d_i^2}{1-d_i^2})_{i=1}^n$.

```
> ols_plot_resid_pot(full_model)
```

- Qualitative Variables as Predictors

The number of qualitative variables required is one less than the number of categories.

For a categorical variable with k categories, we create $k-1$ indicator variables $E_1, \dots, E_{k-1}$ such that $E_i = \begin{cases} 1 & \text{is in } i\text{-th category} \\ 0 & \text{otherwise} \end{cases}$.

(e.g. The model with two categorical variables: $Y = \beta_0 + \alpha_1 E_1 + \alpha_2 E_2 + \alpha_3 E_3 + \gamma_1 F_1 + \gamma_2 F_2$)

```
> data$E_i <- rep(0, n)
> data$E_i[which(data$E == i)] <- 1 # manual approach

> data$E <- as.factor(data$E)
> data <- within(data, E <- relevel(E, ref = k)) # automatic approach
```

- Interaction Variables

If the simple model does not fit well, we add multiplicative terms to the model.

(e.g., $Y = \beta_0 + \alpha_1 E_1 + \alpha_2 E_2 + \alpha_3 E_3 + \gamma_1 F_1 + \gamma_2 F_2$
 $+ \delta_{11} E_1 \cdot F_1 + \delta_{12} E_1 \cdot F_2 + \delta_{21} E_2 \cdot F_1 + \delta_{22} E_2 \cdot F_2 + \delta_{31} E_3 \cdot F_1 + \delta_{32} E_3 \cdot F_2$)

```
> data$E_iF_j <- data$E_i * data$F_j
> model = lm(S ~ ., data = data) # manual approach

> model = lm(S ~ . + E * F, data = data) # automatic approach
```

- Systems of Regression Equations

1. Models with different slopes and different intercepts

Model 1: $Y_1 = \beta_{01} + \beta_{11} X_1 + \varepsilon_1$

Model 2: $Y_2 = \beta_{02} + \beta_{12} X_2 + \varepsilon_2$

which is equivalent to $\begin{cases} H_0 : Y_j = \beta_0 + \beta_1 X_j + \varepsilon_j \\ H_1 : Y_j = \beta_0 + \beta_1 X_j + \gamma Z_j + \delta Z_j X_j + \varepsilon_j \end{cases}, Z_{ij} = \begin{cases} 0 & j = 0 \\ 1 & j = 1 \end{cases}$

We use F -test to test the models.

> **anova**(reduced_model, full_model)

2. Models with the same slopes and different intercepts

Model 1: $Y_1 = \beta_{01} + \beta_1 X_1 + \varepsilon_1$

Model 2: $Y_2 = \beta_{02} + \beta_1 X_2 + \varepsilon_2$

which is equivalent to $\begin{cases} H_0 : Y_j = \beta_0 + \beta_1 X_j + \varepsilon_j \\ H_1 : Y_j = \beta_0 + \beta_1 X_j + \gamma Z_j + \varepsilon_j \end{cases}, Z_j = \begin{cases} 0 & j = 0 \\ 1 & j = 1 \end{cases}$

We use F -test to test the models. Or equivalently, we test $t = \frac{\hat{\gamma}}{\text{s.e.}(\hat{\gamma})} \sim t_{n-3}$

3. Models with different slopes and the same intercepts

Model 1: $Y_1 = \beta_0 + \beta_{11} X_1 + \varepsilon_1$

Model 2: $Y_2 = \beta_0 + \beta_{12} X_2 + \varepsilon_2$

which is equivalent to $\begin{cases} H_0 : Y_j = \beta_0 + \beta_1 X_j + \varepsilon_j \\ H_1 : Y_j = \beta_0 + \beta_1 X_j + \delta Z_j X_j + \varepsilon_j \end{cases}, Z_j = \begin{cases} 0 & j = 0 \\ 1 & j = 1 \end{cases}$

We use F -test to test the models. Or equivalently, we test $t = \frac{\hat{\delta}}{\text{s.e.}(\hat{\delta})} \sim t_{n-3}$

- ANOVA

We wish to compare the means of $k \geq 2$ populations. A random sample $y_{1j}, y_{2j}, \dots, y_{n_j j}$ of size n_j is taken from the j -th population with mean $\mu_j, j = 1, \dots, k$. We have a total of $n = n_1 + \dots + n_k$ observations on the response variable.

$Y = \mu_0 + \mu_1 X_1 + \dots + \mu_{k-1} X_{k-1} + \varepsilon$. The control group becomes $Y = \mu_0 + \varepsilon$.

$\begin{cases} H_0 : \mu_1 = \dots = \mu_k \\ H_1 : \text{not all of them are equal} \end{cases}$

which is equivalent to $\begin{cases} H_0 : \beta_0 + \beta_1 X_1 + \dots + \beta_{k-1} X_{k-1} + \varepsilon \\ H_1 : \beta_0 + \varepsilon \end{cases}$

- Transformation to Achieve Linearity

We need to transform the variables with nonlinear relationships to achieve linearity. A regression model is linear when the parameters in the model occur linearly.

Relationship	Substitution		Linear model
$Y = \alpha X^\beta$	$Y' = \log Y$	$X' = \log X$	$Y' = \log \alpha + \beta X'$
$Y = \alpha e^{\beta X}$	$Y' = \log Y$		$Y' = \log \alpha + \beta X$
$Y = \alpha + \beta \log X$		$X' = \log X$	$Y = \alpha + \beta X'$
$Y = \frac{X}{\alpha X - \beta}$	$Y' = \frac{1}{Y}$	$X' = \frac{1}{X}$	$Y' = \alpha - \beta X$
$Y = \frac{e^{\alpha + \beta X}}{1 + e^{\alpha + \beta X}}$	$Y' = \log \frac{Y}{1 - Y}$		$Y' = \alpha + \beta X$

When $Y = \alpha e^{\beta X} \varepsilon'$, we transform $\ln Y = \beta_0 + \beta_1 X + \varepsilon$. We have $e^{\hat{\beta}_0 - \text{Var}(\hat{\beta}_0)/2}$ is approximately unbiased of estimating α .

However, many nonlinear models cannot be linearized.

- Transformation to Stabilize Variance

We can also transform variables to stabilize the error variance.

Distribution	Variance	Substitution	Variance
$Y \sim \text{Poisson}(\mu)$	$\text{Var}(Y) = \mu$	$Y' = \sqrt{Y}$ or $Y' = \sqrt{Y} + \sqrt{Y+1}$	0.25
$Y \sim \text{Binomial}(n, \frac{\mu}{n})$	$\text{Var}(Y) = \mu(1 - \frac{\mu}{n})$	$\arcsin \sqrt{Y}$	$\frac{0.25}{n}$
$Y \sim \text{NB}$	$\text{Var}(Y) = \mu + \lambda^2 \mu^2$	$Y' = \frac{1}{\lambda} \sinh^{-1}(\lambda \sqrt{Y} [+0.5])$	0.25

- Removal of Heteroscedastic Errors

We hypothesize that $\text{Var}(\varepsilon_i) = k^2 x_i^2$. By replacing $Y' = \frac{Y}{X}$, $X' = \frac{1}{X}$, $\beta'_0 = \beta_0$, $\beta'_1 = \beta_0$, $\varepsilon' = \frac{\varepsilon}{X}$, we get $Y' = \beta'_0 + \beta'_1 X' + \varepsilon'$.

We can also perform Weighted Least Squares (WLS), whose estimates are obtained by minimizing $\sum_{i=1}^n \frac{1}{x_i^2} (y_i - \beta_0 - \beta_1 x_i)^2$, which is equivalent to performing OLS on $Y' = \frac{Y}{X} \sim X' = \frac{1}{X}$.

- Power Transformation

Fit the model $Y^{\lambda_1} \sim X^{\lambda_2}$. Take $\lambda_1, \lambda_2 = \pm 2, \pm 1.5, \pm 1, \pm 0.5, 0$, where $\lambda = 0$ denotes log. We examine the corresponding $\text{Cor}(Y^{\lambda_1}, X^{\lambda_2})$.

- Autocorrelation

Plot the sequence plot and get the sequence of signs of the residuals. A run is a sequence of one sign.

Assume there are n_1 positive residuals and n_2 negative residuals, we should have $n_{\text{run}} \sim \mathcal{N}(\mu, \sigma^2)$, where

$$\mu = \frac{2n_1 n_2}{n_1 + n_2} + 1, \sigma^2 = \frac{2n_1 n_2 (2n_1 n_2 - n_1 - n_2)}{(n_1 + n_2)^2 (n_1 + n_2 - 1)}$$

We can use the run test statistic $\frac{n_{\text{run}} - \mu}{\sigma} \sim \mathcal{N}(0, 1)$ to perform runs test.

Assumption: $\varepsilon_t = \rho \varepsilon_{t-1} + \omega_t$, $|\rho| < 1$, $\omega_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\omega^2)$. We test the hypothesis $\begin{cases} H_0 : \rho = 0 \\ H_1 : \rho > 0 \end{cases}$.

Durbin-Watson Statistic $d = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2}$. We estimate ρ by $\hat{\rho} = \frac{\sum_{t=2}^n e_t e_{t-1}}{\sum_{t=1}^n e_t^2}$. We have $d \doteq 2(1 - \hat{\rho})$

$$d \in \begin{cases} (0, d_L) \implies \text{reject } H_0 \\ (d_L, d_U) \implies \text{the test is inconclusive. The critical values } d_L, d_U \text{ depend on } p \text{ and } n. \\ (d_U, 2) \implies \text{do not reject } H_0 \end{cases}$$

Tests for negative autocorrelation are seldom performed. To perform, one works with $4 - d$ instead of d .

Let $\begin{cases} y_t^* = y_t - \rho y_{t-1} \\ x_t^* = x_t - \rho x_{t-1} \end{cases}$. We perform regression on $Y^* \sim X^*$ and get $\hat{\beta}_0^*, \hat{\beta}_1^*$.

The estimates of the parameters in the original model are $\hat{\beta}_0 = \frac{\hat{\beta}_0^*}{1 - \hat{\rho}}$, $\hat{\beta}_1 = \hat{\beta}_1^*$.

Iterative procedure by Cochrane and Orcutt

Compute $\hat{\beta}_0, \hat{\beta}_1$ by fitting $Y = \beta_0 + \beta_1 X + \varepsilon$. Compute the residuals.

For $k = 1, 2, \dots$

$$\text{Estimate } \hat{\rho}^{(k)} = \frac{\sum_{t=2}^n e_t e_{t-1}}{\sum_{t=1}^n e_t^2}.$$

Let $y_t^{*(k)} = y_t - \hat{\rho}^{(k)} y_{t-1}$ and $x_t^{*(k)} = x_t - \hat{\rho}^{(k)} x_{t-1}$. Fit $Y^{*(k)} \sim X^{*(k)}$ and get $\hat{\beta}_0^{(k)}, \hat{\beta}_1^{(k)}$.

Examine the residuals of $Y = \hat{\beta}_0^{(k)} + \hat{\beta}_1^{(k)} X + \varepsilon$. If still autocorrelated, repeat; else, terminate.

- Collinearity

Typical indications of collinearity

1. High R^2 / Significant F -value but high p -value (low t -value) for the coefficient estimates.
2. High correlation coefficients among predictor variables.
3. Estimated coefficients are contrary to prior expectation.
4. Large changes in the estimated coefficients when a variable is added or deleted.
5. Large changes in the estimated coefficients when a data point is altered or dropped.

Variance Inflation Factor $\text{VIF}_j = \frac{1}{1 - R_j^2}$, $j = 1, \dots, p$, where R_j^2 is the R^2 for $X_j \sim X_1, \dots, X_p$ without X_j .

VIF_j is the factor by which $\text{Var}(\hat{\beta}_j)$ is increased due to collinearity.

> **VIF**(model)

$D^2 = \sum_{j=1}^p E[(\hat{\beta}_j - \beta_j)^2] = \sigma^2 \sum_{j=1}^p VIF_j$ is a measure of precision of the least squares estimators.

$\overline{VIF} = \frac{\sum_{i=1}^p VIF_j}{p}$ is also an index of collinearity.

- Ridge Regression

Let $\tilde{\mathbf{X}} = [\tilde{\mathbf{x}}_1 \ \tilde{\mathbf{x}}_2 \ \dots \ \tilde{\mathbf{x}}_p] \in \mathbb{R}^{n \times p}$, where every $\tilde{\mathbf{x}}_j$ is standardized. $\tilde{\mathbf{y}} = \begin{bmatrix} \tilde{y}_1 \\ \vdots \\ \tilde{y}_n \end{bmatrix} \in \mathbb{R}^{n \times 1}$ is also standardized.

We get the normal equations for ridge regression $\left(\frac{\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}}{n-1} + k \mathbf{I}_p \right) \boldsymbol{\theta} = \frac{\tilde{\mathbf{X}}^\top \tilde{\mathbf{y}}}{n-1}$.

Therefore the estimated ridge regression coefficients $\hat{\boldsymbol{\theta}}(k) = \left(\frac{\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}}{n-1} + k \mathbf{I}_p \right)^{-1} \frac{\tilde{\mathbf{X}}^\top \tilde{\mathbf{y}}}{n-1}$

$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \left(\frac{1}{n-1} \|\tilde{\mathbf{X}}\boldsymbol{\theta} - \tilde{\mathbf{y}}\|^2 + k \|\boldsymbol{\theta}\|^2 \right)$, which penalizes large $\|\boldsymbol{\theta}\|$.

Total Variance(k) = $\sum_{j=1}^p \text{Var}(\hat{\theta}_j(k)) = \sigma^2 \sum_{j=1}^p \frac{\lambda_j}{(\lambda_j + k)^2}$, where λ_j is the j -th largest eigenvalue of $\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}$.

We can plot the $\hat{\theta}_j(k)$ - k curve (ridge trace). If $\hat{\theta}_j$ fluctuates when k approaches zero, collinearity is at work.

After getting $\hat{\boldsymbol{\theta}}$, we need to restore the $\hat{\boldsymbol{\beta}}$ by $\begin{cases} \hat{\beta}_j = \frac{s_y}{s_{x_j}} \hat{\theta}_j & j = 1, \dots, p \\ \hat{\beta}_0 = \bar{y} - \sum_{j=1}^p \hat{\beta}_j \bar{x}_j \end{cases}$.

We have 3 methods to select a value of k .

1. Fixed point method: $k = \frac{p \hat{\sigma}^2(0)}{\sum_{j=1}^p (\hat{\theta}_j(0))^2}$, where $\hat{\sigma}^2(0)$ is the corresponding residual mean square.

2. Iterative method: Start with some initial estimate of k_0 , and preform $k_t = \frac{p \hat{\sigma}^2(0)}{\sum_{j=1}^p (\hat{\theta}_j(k_{t-1}))^2}$ until convergence.

Iterative method may not converge.

3. Ridge trace method: observe the Ridge trace and choose the smallest k when all $\hat{\theta}_j(k)$ are stable.
 $VIF_j(k) < 10$ should also hold.

$VIF_j(k)$ is the j -th diagonal entry of $\left(\frac{\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}}{n-1} + k \mathbf{I}_p \right)^{-1} \frac{\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}}{n-1} \left(\frac{\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}}{n-1} + k \mathbf{I}_p \right)^{-1}$.

```
> library(glmnet)
> X <- as.matrix(data[, 2:p+1])
> y <- as.matrix(data[, 1])
> ridge_model <- glmnet(X, y, alpha = 0)
> summary(ridge_model)
```

- Motivations and Consequences of Variable Selection

Full model: $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \dots + \beta_q X_q + \varepsilon$

Reduced model: $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$

$\begin{cases} H_0 : \text{Full model is adequate} \\ H_1 : \text{Reduced model is adequate} \end{cases}$

Assume $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \dots + \beta_q X_q + \varepsilon$.

Denote the regression parameters fitted to the full model by $\hat{\beta}_0^*, \hat{\beta}_1^*, \dots, \hat{\beta}_q^*$, the reduced model by $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$.

Let $\hat{y}_i = \tilde{\mathbf{x}}_i^\top \hat{\boldsymbol{\beta}}$ and $\hat{y}_i^* = \tilde{\mathbf{x}}_i^\top \hat{\boldsymbol{\beta}}^*$. Let $\mathbf{X}_1$ be the submatrix of $\mathbf{X}$ with p variables.

We have

1. $\text{Var}(\hat{\beta}_j^*) \geq \text{Var}(\hat{\beta}_j)$, $j = 0, 1, \dots, p$
2. $\hat{\beta}_j^*$ is unbiased, while $\hat{\beta}_j$ is biased in general. $(E[\hat{\beta}_j] = \mathbf{e}_j^\top ((\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}} \boldsymbol{\beta}))$
3. $\text{Var}(\hat{y}_i^*) \geq \text{Var}(\hat{y}_i)$, $i = 1, \dots, n$
4. $\hat{y}_i^*$ is unbiased, while $\hat{y}_i$ is biased in general.
5. $\hat{\sigma}^{*2}$ is unbiased, while $\hat{\sigma}^2$ is biased upward ($E[\hat{\sigma}^2] > \sigma^2$).

Definition of the mean square error (MSE): $\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + (E[\hat{\theta}] - \theta)^2$

A regression model can be used as

1. Description and model building: Goal: minimize the number of variables for ease of interpretation.
2. Estimation and prediction: Goal: minimize $\text{MSE}(\hat{y}_{\text{new}})$.
3. Control: Goal: minimize $\text{MSE}(\hat{\boldsymbol{\beta}})$.

- Criteria for Evaluating Equations

We need a criterion to judge the adequacy of various fitted equations.

```
> library(olsrr)
```

1. Residual mean square $\text{RMS}_p = \frac{\text{SSE}_p}{n-p}$. Models with smaller RMS are preferred.
2. $R_p^2 = 1 - (n-p) \frac{\text{RMS}_p}{\text{SST}}$, $R_{ap}^2 = 1 - (n-1) \frac{\text{RMS}_p}{\text{SST}}$. Models with larger R_{ap}^2 are preferred.

```
> summary(reduced_model)$adj.r.squared
```

3. To estimate $J_p = \frac{\sum_{i=1}^n \text{MSE}(\hat{y}_i)}{\sigma^2}$, we use Mallows' $C_p = \frac{\text{SSE}_p}{\hat{\sigma}^2} + (2p - n)$, where $\hat{\sigma}^2 = \frac{\text{SSE}_{\text{FM}}}{n-p-1}$ is obtained from the full model. $E\left[\frac{\text{SSE}_p}{\hat{\sigma}^2}\right] = n-p$ if the reduced model is true. Models with C_p closer to p are preferred.

We can use the C_p - p scatter plot with the line $C_p = p$ to examine the models.

```
> ols_mallows_cp(reduced_model, full_model)
```

4. Akaike information criterion $\text{AIC}_p = n \log\left(\frac{\text{SSE}_p}{n}\right) + 2p$. Models with smaller AIC_p are preferred.
Models with AIC not differing by 2 should be treated as equally adequate.

```
> ols_aic(reduced_model, method = "SAS")
```

5. Bayes information criterion $\text{BIC}_p = n \log\left(\frac{\text{SSE}_p}{n}\right) + p \log n$. Models with smaller BIC_p are preferred.

```
> ols_sbc(reduced_model, method = "SAS")
```

- Variable Selection Procedures

The variables are collinear if $\exists j \text{ VIF}_j > 10$, or $\kappa = \sqrt{\frac{\lambda_{\max}}{\lambda_{\min}}} > 15$, or $\exists j \lambda_j < 0.01$, or $\sum_{j=1}^q \frac{1}{\lambda_j} > 5q$,

where λ_j are the eigenvalues of the correlation matrix. If the variables are non-collinear, we use forward selection procedure, backward elimination procedure and stepwise method for variable selection.

1. Forward selection procedure (FS):

Pick a threshold $\tau > 0$.

For $k = 1, \dots, q$:

Add the variable X_j with the highest $\text{Cor}(X_j, \mathbf{e}_{k-1})$, denoted as X_k . When $k = 1$, $\mathbf{e}_0 = \mathbf{y}$.

Fit the model $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \varepsilon$. Obtain the residuals $\mathbf{e}_k$ and t_j 's.

If $\min_{1 \leq j \leq k} |t_j| < \tau$:
 Stop the process and return the model $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_{k-1} X_{k-1} + \varepsilon$.
 Else:
 Continue.

2. Backward elimination procedure (BS):

Pick a threshold $\tau > 0$.
 For $k = q, \dots, 1$:
 Obtain the t -values t_j 's when fitted in the model $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \varepsilon$.
 If $\min_{1 \leq j \leq k} |t_j| > \tau$.
 Stop the process and return the model $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \varepsilon$.
 Else:
 Drop X_j with the lowest $|t_j|$, say X_k .

For the selected candidate equations, we can use C_p , RMS, AIC, BIC, etc. to evaluate them.

Note that the calculation of C_p needs $\hat{\sigma}^2$, which may be large if the full model contains too many irrelevant variables. Therefore when a good estimate of σ^2 is not available, C_p is of only limited usefulness.

- Logistic Regression

Logit model: $Y = f(X_1, \dots, X_p)$ where $Y \in \{0, 1\}$, $X_j \in \mathbb{R}, j = 1, \dots, p$.

We choose $f(z) = \frac{e^z}{1 + e^z} \in (0, 1)$, called a logistic response function.

If we choose $f(z) = \Phi(z) \in (-1, 1)$, we get the probit model.

$$\pi = P[Y = 1 | X_1 = x_1, \dots, X_p = x_p] = \frac{e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p}}{1 + e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p}} = \frac{\exp(\boldsymbol{\beta}^T \tilde{\mathbf{x}})}{1 + \exp(\boldsymbol{\beta}^T \tilde{\mathbf{x}})}$$

$$1 - \pi = P[Y = 0 | X_1 = x_1, \dots, X_p = x_p] = \frac{1}{1 + e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p}} = \frac{1}{1 + \exp(\boldsymbol{\beta}^T \tilde{\mathbf{x}})}$$

$$\text{Odds ratio: } \frac{\pi}{1 - \pi} = e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p} = \exp(\boldsymbol{\beta}^T \tilde{\mathbf{x}})$$

$$\text{Logit: } \log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

$$\text{Likelihood function } L(\boldsymbol{\beta}) = \prod_{\substack{y_i = 1 \\ 1 \leq i \leq n}} P[y_i = 1 | \mathbf{x}_i] \prod_{\substack{y_i = 0 \\ 1 \leq i \leq n}} P[y_i = 0 | \mathbf{x}_i] = \prod_{i=1}^n \frac{\exp(y_i \boldsymbol{\beta}^T \tilde{\mathbf{x}}_i)}{1 + \exp(\boldsymbol{\beta}^T \tilde{\mathbf{x}}_i)}$$

The maximum likelihood estimator $\hat{\boldsymbol{\beta}} = \arg_{\boldsymbol{\beta}} \max L(\boldsymbol{\beta})$

$$\text{The log-likelihood } l(\boldsymbol{\beta}) = \log L(\boldsymbol{\beta}) = \sum_{i=1}^n \left(y_i \boldsymbol{\beta}^T \tilde{\mathbf{x}}_i - \log(1 + \exp(\boldsymbol{\beta}^T \tilde{\mathbf{x}}_i)) \right)$$

Equivalently, $\hat{\boldsymbol{\beta}} = \arg_{\boldsymbol{\beta}} \max l(\boldsymbol{\beta})$, which has no closed form and requires iterative approach to solve.

```
> model <- glm(Y ~ ., data = data, family = "binomial")
> summary(model)
```

- Model Testing and Inference on Coefficients

Residual deviance $D = -2l(\hat{\boldsymbol{\beta}}) \geq 0$. Saturated model has $D = 0$.

Null deviance $D_0 = -2l(\hat{\beta}_0 \mathbf{e}_1)$, which is the residual deviance of the worst model, with $\beta_1 = \dots = \beta_p = 0$.

The R^2 statistic $R^2 = 1 - \frac{D}{D_0}$

$$\text{Hypothesis testing: } \begin{cases} H_0 : \pi = \frac{e^{\beta_0}}{1 + e^{\beta_0}} \\ H_1 : \pi = \frac{e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p}}{1 + e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p}} \end{cases}$$

G -statistic $G = D_0 - D$ is the counterpart of F -statistic in linear regression. $G \sim \chi_p^2$ under H_0 .

$E[\hat{\beta}_j] \doteq \beta_j$ is approximately unbiased.

$$\beta_j \sim \mathcal{N}, \text{Cov}(\beta_i, \beta_j) = v_{ij}, \text{ where } \mathbf{V} = (v_{ij})_{i,j=0}^p = \left(\mathbf{\tilde{X}}^\top \text{diag} \left(\frac{\exp(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i)}{(1 + \exp(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i))^2} \right)_{i=1}^n \mathbf{\tilde{X}} \right)^{-1}.$$

$$\text{Wald statistic } \frac{\hat{\beta}_j - \beta_j}{\text{s.e.}(\hat{\beta}_j)} \sim \mathcal{N}(0,1) \quad \text{The } 1 - \alpha \text{ confidence interval for } \beta_j: \hat{\beta}_j \pm z_{\frac{\alpha}{2}} \cdot \text{s.e.}(\hat{\beta}_j)$$

$$\text{The } 1 - \alpha \text{ confidence interval for } e^{\beta_j} \text{ (odds ratio): } \exp \left(\hat{\beta}_j \pm z_{\frac{\alpha}{2}} \cdot \text{s.e.}(\hat{\beta}_j) \right)$$

- Logistic Regression Diagnostic

Counterparts of e_i in logistic regression:

$$\begin{aligned} 1. \quad \text{Standardized deviance residual } \text{DR}_i &= \begin{cases} \sqrt{-2 \ln(\hat{\pi}_i)} & y_i = 1 \\ -\sqrt{-2 \ln(1 - \hat{\pi}_i)} & y_i = 0 \end{cases} \\ 2. \quad \text{Standardized Personian residual } \text{PR}_i &= \frac{y_i - \hat{\pi}_i}{\sqrt{\hat{\pi}_i(1 - \hat{\pi}_i)}} \end{aligned}$$

We can examine the scatter plot of $(\hat{\pi}_i, \text{DR}_i)_{i=1}^n, (\hat{\pi}_i, \text{PR}_i)_{i=1}^n, (\text{DR}_i, i)_{i=1}^n, (\text{PR}_i, i)_{i=1}^n$.

- Determination of Variables to Retain

$$\begin{cases} H_0 : \text{Reduced model } \pi = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)}} \text{ is adequate} \\ H_1 : \text{Full model } \pi = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_{p+q} x_{p+q})}} \text{ is adequate} \end{cases}$$

We have $2(l_{p+q} - l_p) \sim \chi_q^2$, where l_p, l_{p+q} are the log-likelihoods of two models.

$$\text{AIC}_p = -2l_p + 2p, \text{BIC}_p = -2l_p + p \log n$$

- Judging the Fit

We judge the fit of the model by calculating the accuracy of classification.

A sample of size n with n_1 observations being $Y = 0$ and n_2 being $Y = 1$, we take $\max \left\{ \frac{n_1}{n}, \frac{n_2}{n} \right\}$ as a base level.

For every pair $((\mathbf{x}_1, y_1 = 0), (\mathbf{x}_2, y_2 = 1)) \in \left\{ (\mathbf{x}_{(i)}, y_{(i)} = 0) \right\}_{i=1}^{n_1} \times \left\{ (\mathbf{x}_{(j)}, y_{(j)} = 1) \right\}_{j=1}^{n_2}$, if

$\hat{\pi}_1 = P[Y = 1 | \mathbf{x}_1] < \hat{\pi}_2 = P[Y = 1 | \mathbf{x}_2]$, we say this pair is correctly classified.

$$\text{Concordance index} = \frac{n_{\text{correctly classified}}}{n_1 n_2}$$

- Multinomial (Polytomous) Logistic Regression

Multinomial logit model: $Y = f(X_1, \dots, X_p)$ where $Y \in \{1, \dots, k\}$

For variables $X_1, \dots, X_p$, choose X_k as the base level.

The logits are $\log \left(\frac{\pi_j(\mathbf{x}_i)}{\pi_k(\mathbf{x}_i)} \right) = \beta_j^\top \tilde{\mathbf{x}}_i, i = 1, \dots, n, j = 1, \dots, p, j \neq k$.

$$\log(\pi_j(\mathbf{x}_i)) = \frac{\exp(\beta_j^\top \tilde{\mathbf{x}}_i)}{1 + \sum_{j \neq k} \exp(\beta_j^\top \tilde{\mathbf{x}}_i)}, j = 1, \dots, p$$